

fixed_h2_pipe_row_parity_gate

fixed_h2_pipe_row_parity_gate.sh

Revision: 2 Last modified: 2026-07-29T00:00:00Z **Gate ID:** CM-FIXED-SUMMARY-ITEM-VISIBILITY (was CM-FIXED-H2-PIPE-ROW-PARITY)

Overview

§11.4.135 standing regression guard for the §11.4.90 / §11.4.91 / §11.4.53 docs-tooling drift that hid **HXC-044** (Bug | Obsolete (→ Fixed.md)) from docs/Fixed_Summary.md.

docs/Fixed.md is MIXED: a pipe table (| Closure | Title | Type | Status | Round | Commit(s) | Evidence |) **AND** H2 detail sections (## HXC/ATM-NNN – ...).

The file name is now historical. It names the *mechanism* this guard used before the 2026-07-29 reconciliation below, not what it asserts. The path is retained deliberately — docs/Issues.md, docs/Issues_Summary.md and this document reference it, and tracker-doc regeneration is operator-deferred, so a rename would strand those references. Renaming is tracked as follow-up, never done silently (§11.4.124).

§11.4.120 reconciliation (2026-07-29, HXC-161)

Original invariant. The guard asserted “every H2 closure heading has a matching pipe-table row” as a **proxy** for what actually mattered — *no closed item is missing from Fixed_Summary.md*. Under the then-canonical generator scripts/generate_fixed_summary.sh, which read **only** the pipe table, the proxy was sound: having a pipe row was exactly what made an item visible.

What changed — the consumer, not the data. Commit 8494380a (2026-07-29):

- scripts/generate_fixed_summary.sh is **superseded and refuses to run** (exits 2 in every mode, writes nothing) after a hand invocation at 0a4df699 rewrote the closed-item tally 344 → 188, dropping 156 items of tracked state.
- docs/Fixed_Summary.md is now **derived from docs/workable_items.db** by the constitution submodule’s Go exporter (renderFixedSummary, export.go), which selects every item with `current_location ==`

'Fixed' and is **representation-agnostic** — a section item and a table item land in the summary identically.

Measured consequence (HEAD bbd236c9): 56 H2 closure sections have no pipe row, and **all 56 are present in Fixed_Summary.md**. The invisibility the proxy existed to prevent no longer occurs; the proxy had become a standing 57-failure red asserting an obsolete era’s mechanism. Per §11.4.120 the gate was **reconciled** to the new mechanism — never fake-passed, never reverted-against.

Why this is a strengthening, not a loosening. Against 0a4df699, the real data-loss commit (summary corrupted down to 20 lines):

	summary-visibility failures reported
retired pipe-row invariant	0 — it never noticed (it exited 1, but only on pipe-row noise: a 57-failure standing red is exactly how a real failure gets masked)
reconciled invariant	246 closed items flagged missing

What it asserts

- **(A)** every docs/Fixed.md H2 closure heading (## HXC-NNN / ## ATM-NNN) appears in docs/Fixed_Summary.md. **Hard FAIL.**
- **(B)** every docs/Fixed.md pipe-table row whose Title cell carries a ticket id appears in docs/Fixed_Summary.md. **Hard FAIL.**
- **(C)** every Obsolete (→ Fixed.md) pipe-table item appears in docs/Fixed_Summary.md. **Hard FAIL.** A strict subset of (B), retained as an explicitly-named check because it is the original HXC-044 forensic anchor, so an Obsolete-specific regression is reported as such.

Fixed.md and Fixed_Summary.md are rendered by **different code paths** (a doc_segments walk vs. renderFixedSummary’s item select), so agreement between them is a real, falsifiable cross-check — not a tautology, as 0a4df699 proved by breaking it.

Honest boundary (§11.4.6 / §11.4.118)

- Ticket-id matching is **bounded** (non-alphanumeric on both sides) so ATM-97 is not spuriously satisfied by ATM-970.
- Legacy pipe rows whose Title cell carries **no ticket id** (the 2026-05-19 i18n-round rows, Tracker HTML + PDF exports per §11.4.19, ...) cannot be

keyed by id and are **out of scope** for (B). Their count is **printed on every run** so the exclusion is visible, not hidden — 87 at HEAD.

- This is a **document-parity** guard (`Fixed.md` ↔ `Fixed_Summary.md`) and reads no database. It proves the two rendered documents agree, **not** that either matches the SQLite SSoT — that `md↔db` direction belongs to CM-WORKABLE-ITEMS-MD-DB-IN-SYNC and CM-SUMMARY-SYNC, both still required.
- The invariant is **architecture-current, not retroactively green**: it also fails against `a665b02d`, because that era's summary was aggregate-counts-only and listed exactly one item id. That is a true property of the old document shape, not a false positive. The invariant became satisfiable at `8494380a`, when the summary gained its DB-derived per-item table.
- An **anti-tautology tripwire** refuses a vacuous PASS: if the extractors match 0 H2 sections *and* 0 ticket-id pipe rows (format drift), the gate FAILs rather than reporting a green it did not earn.

Prerequisites

`awk`, `grep`, `mktemp`, `bash`. No network, no credentials, no database.

Usage

```
scripts/gates/fixed_h2_pipe_row_parity_gate.sh
# RED-baseline (§11.4.115) – reproduce the defect on temp copies
    and assert the
# guard genuinely catches it (exit 0 = guard is not blind):
RED_MODE=1 scripts/gates/fixed_h2_pipe_row_parity_gate.sh
```

`FIXED_PARITY_STRICT` is **gone** — it toggled the retired pipe-row invariant between WARN and FAIL. All three current invariants are unconditional hard FAILs.

Edge cases

- `RED_MODE=1` operates on copies under `$TMPDIR` (cleaned on `EXIT`); it never mutates the live `docs/Fixed.md` / `docs/Fixed_Summary.md`.
- `RED_MODE=1` deletes **two** victims covering **both** representations — `HXC-044` (Obsolete, has a pipe row → trips (B)+(C)) and `HXC-157` (section-only, no pipe row → trips (A)). `HXC-157` is the load-bearing one: under the retired invariant its disappearance from the summary was completely invisible. A future edit that silently narrows the guard to one representation is caught here.

Internal behaviour

Parses pipe data rows by column (col1 = Closure date, col2 = Title, col3 = Type, col4 = Status, ...; awk fields shift by one because each line starts with |), extracts the leading (HXC|ATM) -<id> token from Title cells and H2 headings, and bounded-matches each id against docs/Fixed_Summary.md.

Related scripts

- constitution/scripts/workable-items (renderFixedSummary in export.go) — the canonical DB-derived generator this guard now protects.
- scripts/generate_fixed_summary.sh — the **superseded** markdown-derived generator (refuses to run; retirement tracked as HXC-160).
- scripts/gates/summary_sync_gate.sh, scripts/gates/workable_items_sync_gate.sh — the md↔db direction this guard deliberately does not cover.
- scripts/gates/obsolete_details_gate.sh, scripts/gates/obsolete_colorize.sh — §11.4.90 sibling gates.

Last verified

2026-07-29 (HEAD bbd236c9) — GREEN (exit 0): 153 H2 sections, 99 ticket-id pipe rows, 1 Obsolete item all present in Fixed_Summary.md; 87 legacy rows reported out of scope. RED_MODE=1 exit 0 (guard correctly FAILs on the broken artifact). Mutation matrix, all exit 1: delete a section-only closure → (A) fires; delete a pipe-row closure → (A)+(B)+(C) fire; empty the summary → 253 failures; empty Fixed.md → anti-tautology tripwire fires.